
KENYA NATIONAL EXAMINATIONS

FORM 4 MATHEMATICS - CAT

TIME: 1 1/2 HOURS

TOTAL MARKS: 30

Name: _____ Adm No: _____ Class: _____

INSTRUCTIONS:

1. Answer ALL questions.
 2. Show all working in the spaces provided.
 3. Marks are awarded for clear working.
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Question 1 (3 marks)

Solve the quadratic equation $3x^2 - 7x + 2 = 0$, giving your answers correct to 2 decimal places.

Question 2 (4 marks)

Find the range of values of x for which $(2x - 1)(x + 3) > 0$. Show all working on a number line.

Question 3 (4 marks)

Given that $\log_2(x + 3) + \log_2(x - 1) = 5$, find the value of x .

Question 4 (3 marks)

Find the equation of a straight line passing through $(-2, 5)$ and perpendicular to the line $2x - 3y + 6 = 0$.

Question 5 (4 marks)

A triangle has vertices $A(1, 2)$, $B(5, 2)$ and $C(5, 6)$. Find:

- (a) The length of AC . [2 marks]

(b) The midpoint M of AC, and show that BM is perpendicular to AC. [2 marks]

Question 6 (4 marks)

Differentiate with respect to x:

(a) $f(x) = 3x^4 - 5x^2 + 7x - 2$ [2 marks]

(b) $g(x) = (2x + 1)^3$ [2 marks]

Question 7 (4 marks)

The table below shows marks scored by 10 students in a test:

Marks: 12, 18, 15, 20, 14, 19, 17, 13, 16, 21

(a) Calculate the mean. [1 mark]

(b) Calculate the standard deviation. [3 marks]

Question 8 (4 marks)

Solve the equation $2\sin^2(\theta) - \sin(\theta) - 1 = 0$ for $0^\circ \leq \theta \leq 360^\circ$.

--- End of Paper ---